ECE 25500: LTspice Assignment IV

IC Cascode Amplifier Design

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Due on: Nov. 15th, 2019 *by* 5:00 PM

Note: Carefully follow the instruction listed in the *LTspice Assignment Guidelines* document to submit your work. Late submissions will **not** be accepted. Please email the course GTA at `rchatrie@purdue.edu` if you have any questions about this assignment. It will be worth 2% of your total grade. ***Important:*** For this assignment, exceptionally, we will **not** respond directly to emails that inquire about anything other than instruction clarification or the logistics of the assignment.

Single-Stage Cascode Amplifier

One of the governing constraints usually imposed an any analog design relates to its frequency response. You'll be introduced to this very important subject last in this course. A thorough treatment is left to a more advanced course. However, even at this current stage, you should realize that designing an audio amplifier (audio frequencies sit between 20 Hz and 20,000 Hz) and designing a WiFi signal transmission amplifier (2.4 GHz) are two very different tasks, which each require very specific design considerations. As it turns out, the cascode configuration configuration is well suited for analog circuits in which high frequency and low noise operation is required. Even though neither of these two aspects of analog design will be directly covered in this course, you should know why the cascode configuration is useful and how to design it at a basic level.

The cascode configuration is best viewed as a multi-stage configuration with a common source stage followed by a common gate stage. The common source is the transconductance amplification stage, and the common gate serves as a current follower. The cascode stage thus provides high gain and a high output impedance. However, in order to leverage these characteristics, one must pay special attention to the bias circuitry. In this assignment, you'll be designing a single-stage cascode amplifier biased with a cascode current mirror (see **Fig.1**).

As an added touch of realism, you'll be using **short-channel** devices to simulate this circuit in LTspice. Specifically, you'll have to tell LTspice to use the 45 nm technology model available on ASU's webpage (see the last section). The MOSFET IC manufacturing process allows one to precisely set the physical dimensions (W and L) of a device. This in turn allows us to set circuit behavior through a careful selection of these dimensions. In this assignment, you'll be doing just that. Note that this exercise is very different from discrete circuit design (LTspice 3).

Design Constraints

Your task is to design the single-stage cascode amplifier shown in **Fig.1** below such that the condition below are satisfied. In this case, you'll have to choose values for the channel width and length of each of the MOSFETs M_1 through M_6 in the circuit (in total, you'll have to choose the value of 12 parameters). The model to use for the MOSFETs will be specified in the last section section. The submission details are provided in the very next section.

(a) The NMOSs and PMOSs must be modelled by ASU's 45 nm device models (see the last section).

(b) You must use the exact circuit topology shown in **Fig.1** below.

(c) The magnitude of the gain is at least 100.

(d) The output DC voltage at node v_{out} is within 10 mV of 0 V.

(e) $V_{DD} = V_{SS} = 1.5$ V.

(f) $R_{REF} = 20$ kΩ.

(g) The overall power consumption is less than 200 µW.

(h) The bias current in each of the two branches is at least 1 µA.

(i) The channel width to length ratio for each of the transistor W/L is at least 1.

(j) The channel lengths L for each of the transistors are non-zero multiples of the minimum channel length 45 nm.

(k) You can add any desired DC offset to v_{in} in order to correctly bias transistor M_1.

Submission

As for the previous LTspice assignment, your submission will consist of a single `.txt` file that contains your chosen values for each of the 13 parameters. Name the file with your Purdue username. Write the values in the file, one line for each, according to the following sequence (use units of nm): $W_1, L_1, W_2, L_2, W_3, L_3, \ldots, W_6, L_6$.

Importing the MOSFET Models

The model instances are available at `http://ptm.asu.edu`. Click on **Nano-CMOS**, which appears on the left panel. Select either NMOS or PMOS, make sure the technology is 45 nm, and click **go**. Leave all parameters at their default value and click **submit** again. Click on **nominal model** to load the model. Repeat these steps for the PMOS model. Save these models into `.txt` files in the same directory as your schematic for this project (for convenience, but you can save them wherever). Open each of them in edit mode, and observe the first non-commented line:

```
.model pmos pmos level = 54
```

This is the same `.model` directive you have seen before. In order to use the model instance correctly, you need give a new name to your model. This means directly changing the file as follows:

```
.model pmos45nm pmos level = 54
```

Save the change and do the same for the NMOS model instance. In order to use the model instances from this point, you need to first import them into the netlist of your circuit, and second, change the `value` parameter of the transistor instances in the netlist to match the model name you wrote in the files (i.e. `pmos45nm` as above). In order to import the files into the netlist, use the following SPICE directive:

```
.lib <path/to/model/instances>
```

Provide an absolute path to make sure the simulator does not default to the default MOSFET models. Again, verify that every transistor instance is using the desired models. Note that you can still customize select parameters such as length and width by right-clicking on the transistor instances.

Important: You must use the **nmos4** and **pmos4** devices in LTspice in order to be able to change the length and width easily. When instantiating these devices, make sure to connect its fourth terminal, the body contact, to its source terminal in all cases.

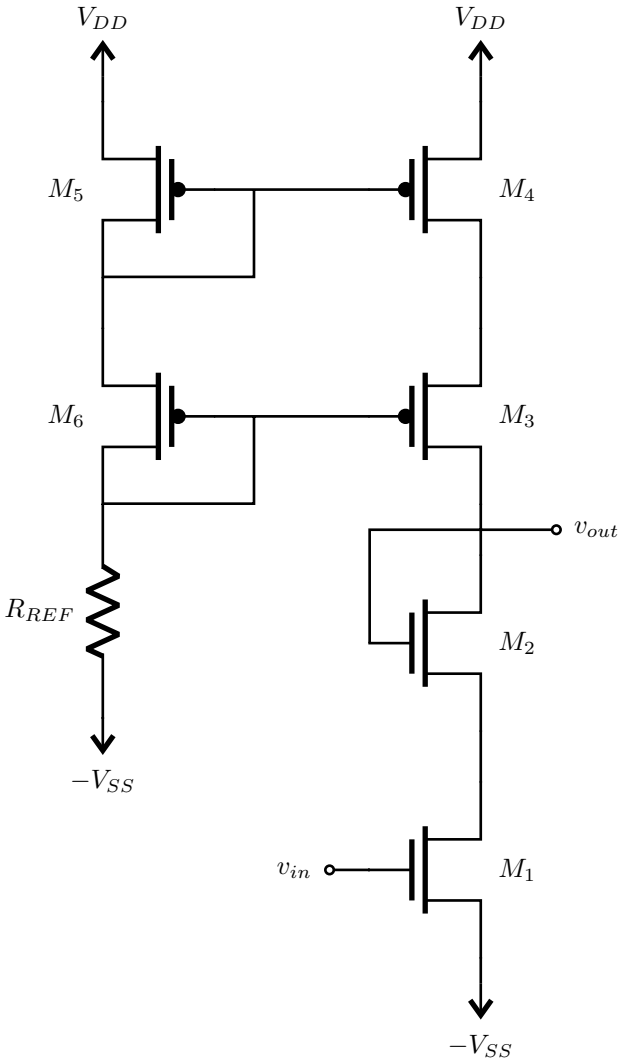


Fig.1 - CMOS Single-stage Cascode Amplifier.